Tutorial

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1 The Cox Regression model 7

```
# get packages

library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
library(survival)
library(knitr)
library(tibble)
library(lubridate)
```

```
Attaching package: 'lubridate'

The following objects are masked from 'package:base':

    date, intersect, setdiff, union
```

```
library(ggsurvfit)
```

```
Loading required package: ggplot2
```

```r
library(gtsummary)
library(tidycmprsk)
```

```
Attaching package: 'tidycmprsk'
```

```
The following object is masked from 'package:gtsummary':

    trial
```

```r
head(lung)
```

inst	time	status	age	sex	ph.ecog	ph.karno	pat.karno	meal.cal	wt.loss
3	306	2	74	1	1	90	100	1175	NA
3	455	2	68	1	0	90	90	1225	15
3	1010	1	56	1	0	90	90	NA	15
5	210	2	57	1	1	90	60	1150	11
1	883	2	60	1	0	100	90	NA	0
12	1022	1	74	1	1	50	80	513	0

Status: 1 –> cencored , 2 –> dead

time: observed survival time in days

Sex : 1 = Male, 2 = Female

```r
# to change to known format

lung <- lung |>
  mutate(
    status = recode(status, "1" = 0, "2" = 1)
  )
```

```r
head(lung[, c("time", "status", "sex")])
```

time	status	sex
306	1	1
455	1	1
1010	0	1
210	1	1
883	1	1
1022	0	1

```r
date_ex <-
  tibble(
    sx_date = c("2007-06-22", "2004-02-13", "2010-10-27"),
    last_fup_date = c("2017-04-15", "2018-07-04", "2016-10-31")
  )

date_ex
```

sx_date	last_fup_date
2007-06-22	2017-04-15
2004-02-13	2018-07-04
2010-10-27	2016-10-31

```r
date_ex <- date_ex |>
  mutate(
    sx_date = ymd(sx_date),
    last_fup_date = ymd(last_fup_date)
  )

date_ex
```

sx_date	last_fup_date
2007-06-22	2017-04-15
2004-02-13	2018-07-04
2010-10-27	2016-10-31

```r
date_ex <- date_ex |>
  mutate(
    os_years = as.duration(sx_date %--% last_fup_date)/ dyears(1)
  )

date_ex
```

sx_date	last_fup_date	os_years
2007-06-22	2017-04-15	9.815195
2004-02-13	2018-07-04	14.387406
2010-10-27	2016-10-31	6.012320

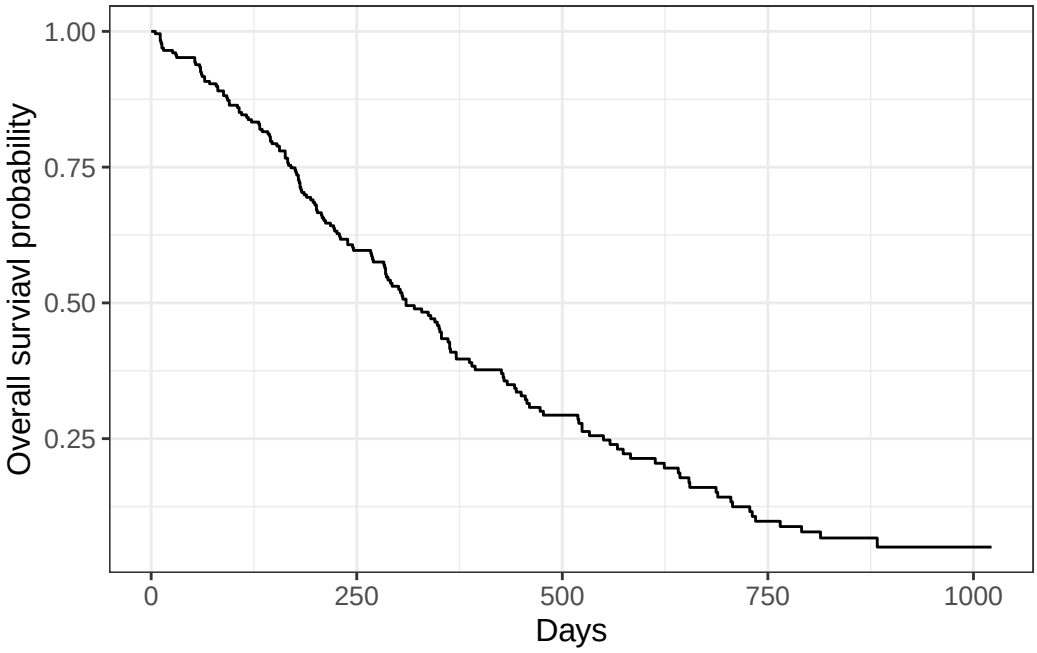
```r
Surv(lung$time, lung$status)[1:10]
```

```
 [1]  306   455  1010+  210   883  1022+  310   361   218   166
```

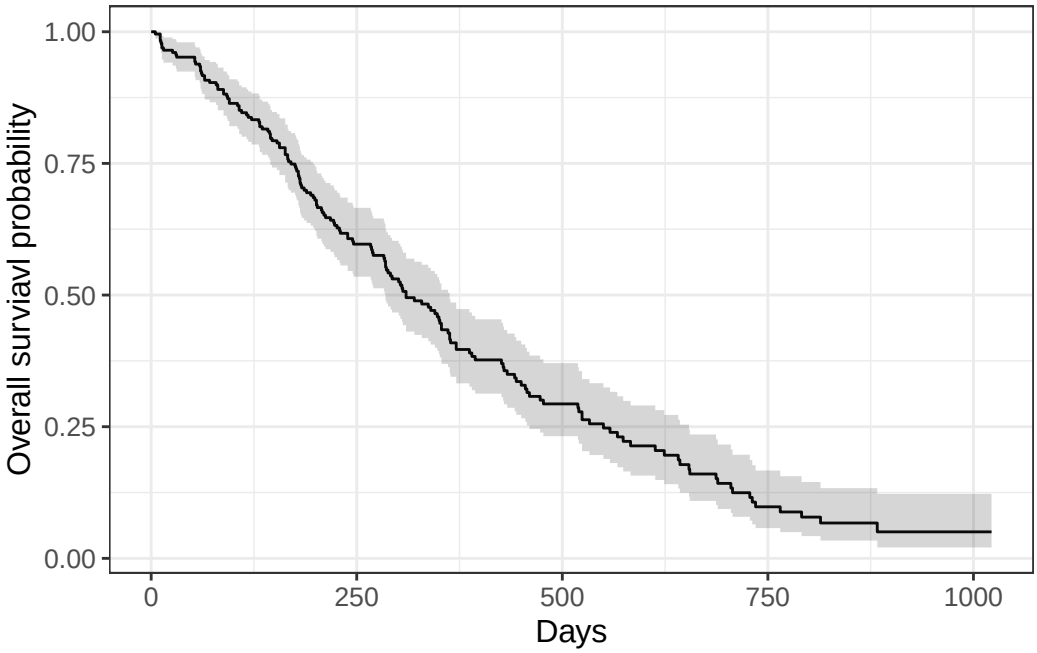
```r
s1 <- survfit(Surv(lung$time, lung$status) ~ 1, data = lung )
str(s1)
```

```
List of 17
 $ n        : int 228
 $ time     : num [1:186] 5 11 12 13 15 26 30 31 53 54 ...
 $ n.risk   : num [1:186] 228 227 224 223 221 220 219 218 217 215 ...
 $ n.event  : num [1:186] 1 3 1 2 1 1 1 1 2 1 ...
 $ n.censor : num [1:186] 0 0 0 0 0 0 0 0 0 0 ...
 $ surv     : num [1:186] 0.996 0.982 0.978 0.969 0.965 ...
 $ std.err  : num [1:186] 0.0044 0.00885 0.00992 0.01179 0.01263 ...
 $ cumhaz   : num [1:186] 0.00439 0.0176 0.02207 0.03103 0.03556 ...
 $ std.chaz : num [1:186] 0.00439 0.0088 0.00987 0.01173 0.01257 ...
 $ type     : chr "right"
 $ logse    : logi TRUE
 $ conf.int : num 0.95
 $ conf.type: chr "log"
 $ lower    : num [1:186] 0.987 0.966 0.959 0.947 0.941 ...
 $ upper    : num [1:186] 1 1 0.997 0.992 0.989 ...
 $ t0       : num 0
 $ call     : language survfit(formula = Surv(lung$time, lung$status) ~ 1, data = lung)
 - attr(*, "class")= chr "survfit"
```

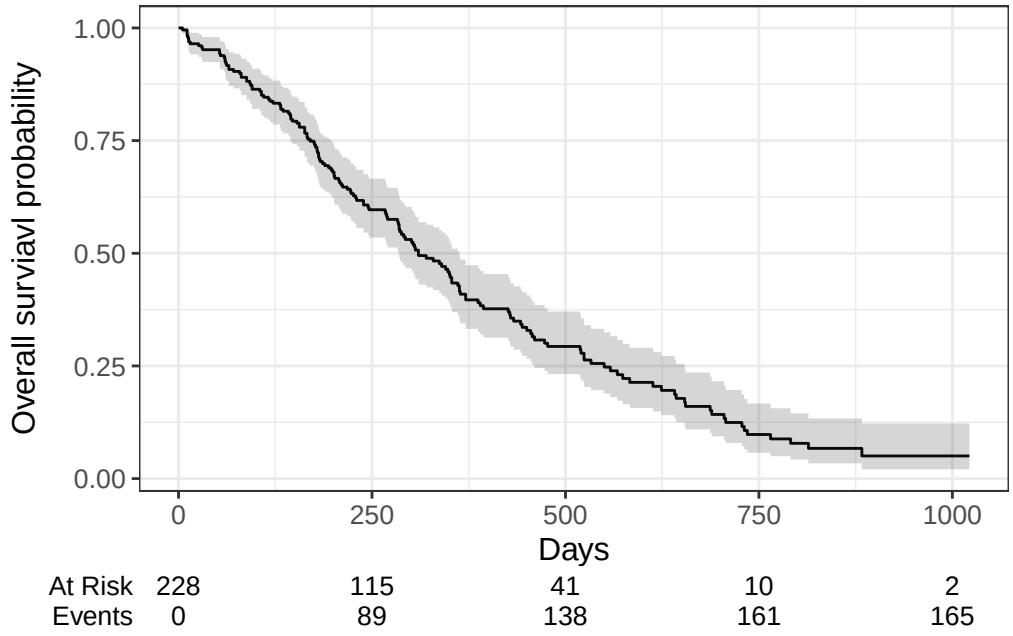
```r
survfit2(Surv(time, status) ~ 1, data = lung) |>
  ggsurvfit() +
  labs(
    x = "Days",
    y = "Overall surviavl probability"
  )
```




```
survfit2(Surv(time, status) ~ 1, data = lung) |>
  ggsurvfit() +
  labs(
    x = "Days",
    y = "Overall surviavl probability"
  ) + add_confidence_interval()
```



```
survfit2(Surv(time, status) ~ 1, data = lung) |>
  ggsurvfit() +
  labs(
    x = "Days",
    y = "Overall surviavl probability"
  ) + add_confidence_interval() +
  add_risktable()
```




```
summary(s1, times = 365.25)
```

```
Call: survfit(formula = Surv(lung$time, lung$status) ~ 1, data = lung)

 time n.risk n.event survival std.err lower 95% CI upper 95% CI
  365     65     121    0.409  0.0358        0.345        0.486
```

From the above, the probability of survival for one years for subjects in the study is 40%.

```
s1 |>
  tbl_survfit(
    times = 365.25,
    label_header = "**1-year survival (95% CI)**"
  )
```

```
s1
```

Characteristic	1-year survival (95% CI)
Overall	41% (34%, 49%)

Characteristic	Median survival (95% CI)
Overall	310 (285, 363)

```
Call: survfit(formula = Surv(lung$time, lung$status) ~ 1, data = lung)

       n events median 0.95LCL 0.95UCL
[1,] 228    165    310     285     363
```

```
s1 |>
  tbl_survfit(
    probs = 0.5,
    label_header = "**Median survival (95% CI)**"
  )
```

```
survdiff(Surv(time, status) ~ sex, data = lung)
```

```
Call:
survdiff(formula = Surv(time, status) ~ sex, data = lung)

        N Observed Expected (O-E)^2/E (O-E)^2/V
sex=1 138      112     91.6      4.55      10.3
sex=2  90       53     73.4      5.68      10.3

 Chisq= 10.3  on 1 degrees of freedom, p= 0.001
```

$$H_0 : S_A(t) = S_B(t)$$

$$H_1 : S_A(t) \neq S_B(t)$$

P-value is 0.001 which is less than 5%. We reject the null hypothesis. We conclude there is significance survival difference between male and female patients.

1 The Cox Regression model

$$h(t|X_i) = h_0(t) \exp(\beta_1 X_{i1} + \cdots + \beta_p X_{ip})$$

Characteristic	HR	95% CI	p-value
sex	0.59	0.42, 0.82	0.001

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

```
cox <- coxph(Surv(time, status) ~ sex, data = lung)
cox
```

```
Call:
coxph(formula = Surv(time, status) ~ sex, data = lung)

       coef exp(coef) se(coef)      z       p
sex -0.5310    0.5880   0.1672 -3.176 0.00149

Likelihood ratio test=10.63  on 1 df, p=0.001111
n= 228, number of events= 165
```

From the above $h(t|X_i) = h_0(t)\exp(-0.5310X_1)$

```
cox |>
  tbl_regression(exp = TRUE)
```

The quantity of interest from a Cox regression model is a **hazard ratio (HR)**. The HR represents the ratio of hazards between two groups at any particular point in time. The HR is interpreted as the instantaneous rate of occurrence of the event of interest in those who are still at risk for the event. It is **not** a risk, though it is commonly mis-interpreted as such. If you have a regression parameter , then HR = exp( ).

A HR < 1 indicates reduced hazard of death whereas a HR > 1 indicates an increased hazard of death.

So the HR = 0.59 implies that 0.59 times as many females are dying as males, at any given time. Stated differently, females have a significantly lower hazard of death than males in these data.